

Principles of Aquaculture Engineering

A. Introduction

A. Aquaculture in the Philippines

- In 2024, the Philippines has a total volume of production of 1,020,823 metric tons of fisheries products with seaweed as the number one aquaculture species in terms of volume of production.
- The fisheries sector in the country relies in three different outputs:

Municipal Fisheries	Refers to the artisanal, subsistence, and small scale fishery that uses vessel 3 gross tons or smaller
Commercial Fisheries	large scale fishery operation using fishing boats with a capacity of more than 3 GT
Aquaculture	Cultivation of aquatic animals and plants in controlled environments

How is aquaculture different from fishing?

Aquaculture basically is fish farming. Aquatic organisms are deliberately cultivated and raised in controlled environments for human consumption. Fishing is more comparable to hunting, where aquatic animals are collected from their natural habitat for human consumption.

Why is aquaculture an important sector?

- to increase food production (combat increasing demand)
- to avoid overharvesting of natural stocks
- to foster economic development through export and employment

B. Types and Classifications of Aquaculture

A. Based on Growing Medium / Source of Water

1. **Freshwater Aquaculture** - aquatic organisms are grown in freshwater sourced from springs, rivers, lakes, and also from developed underground water resources. Species such as tilapia, carp, milkfish, catfish, and freshwater shrimp are typically cultured in pens, cages, or ponds. (<0.5 ppt)
2. **Brackish Water Aquaculture** - brackish water refers to the combination of fresh and sea water, typically found in estuaries and mangrove areas where the river mixes with tidal water. The salinity of brackish water is often variable depending on the tidal phase and the freshwater discharge. The typical species cultured in brackish water are milkfish, tilapia, crabs, prawns, etc. (0.5 - 35 ppt)
3. **Mariculture (Sea Water Aquaculture)** - this refers to the use of sea water as the growth medium, typically in coastal areas. Fish species such as lapu-lapu and kitang are cultured in pens and cages while other aquaculture species such as seaweeds,oysters, and mussels are cultured in shallow sandy areas. (>35 ppt)

B. Based on Production Methods

1. Extensive Method - this is characterized by minimal stocking density, the use of natural feed, and minimal water exchange. (1-3 fish per sq. m)
2. Semi-Intensive Method - this is characterized by moderate stocking density, the use of natural and supplemental feed (with fertilizer), and with partial water exchange. (4-6 fish per sq. m)
3. Intensive Method - this is characterized hy high stocking density, the use of artificial feeds, and high water exchange rate (for maximum aeration). (6 and more fish per sq. m)

C. Based on Aquaculture Products

1. Nutraceutical Products - aquatic organisms are raised for the purpose of food and medicine.
2. Recreation - aquatic organisms are raised primarily for leisure (bait fish for sport fishing)
3. Ornamental - aquatic organisms are raised for their aesthetic appeal.

D. Based on Water Use

1. Open Systems - aquatic organisms are raised in natural or man-made bodies of water (such as in cages, pens, and enclosures)
2. Flow Through Systems - the water passes through the aquaculture system and is then discharged
3. Closed Systems - the water is reconditioned and then recirculated throughout the system.

E. Based on Purpose or Function

1. Broodstock - enclosures for the conditioning of mature fish for breeding.
2. Hatchery -
3. Nursery -
4. Growout / rearing ponds -
5. Combination -

F. Based on Number of Species

1. Monoculture - only one species if cultured.
2. Polyculture - two or more species are cultured in one system.

Stenohaline Organism - can withstand small variations in the salinity of the growing medium. Example is carp and mackerel.

Euryhaline Organism - can withstand a wide variation in salinity of the growing medium such as in brackish water such as prawns and crabs.

Anadromous Fish - spawns in freshwater and then spends their lives in salt water habitats.

Katadromous Fish - spawns in salt water and then spends their lives in fresh water habitats.

C. Ecological Considerations in Selecting Aquaculture Site

A. Water Supply

The availability and quality of water supply is the most crucial factor to be considered in aquaculture. An adequate supply of water free from any form of pollution should be available throughout the year (or at least the whole growing period).

Minimum requirement of 5 L/sec/ha of pond for systems supplied with water from reservoirs such as rivers, wells, groundwater.

Minimum requirement of 10-15 ha of catchment area to 1 ha of pond for systems supplied with rainwater.

B. Water Quality

Physical Characteristics

Temperature	Temperature affects the metabolic rate, growth, reproduction of aquatic animals as their body temperature is regulated by the surrounding temperature (ectothermic).
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Color	Refers to the visible hue or tint of the water due to dissolved substances.
Turbidity	Refers to the clarity of the water as an effect of the scattering of light due to suspended particles.
Suspended Solids	Refers to small solid particles that are dispersed in water.

Chemical Characteristics

pH	
Salinity	
Dissolved Oxygen	
Dissolved Solids	
Ammonia	

Acceptable range of water quality parameters for aquaculture applications

<u>Parameter</u>	<u>Acceptable range</u>
pH	7.0 - 8.5
Alkalinity (as CaCO₃)	100 - 400 ppm or mg/L
Dissolved oxygen	> 3 mg/L
Total hardness	100 - 400
Carbon dioxide	0 - 15
Iron – ferrous (Fe²⁺)	Less than 0.1 ppm
Unionized Ammonia (NH₃)	Less than 0.02 ppm
Nitrate-N (NO₃-N)	0 - 3.0 ppm
Nitrite-N (NO₂-N)	0.1 ppm

C. Type of Soil

The soil should not be highly permeable to prevent the excessive seepage of water from the pond. It should also be able to support vegetation growth at the pond bottom.

Clayey soils have good compaction and low permeability. As the clay content decreases, suitability for ponds also decreases.

Sandy clay to clayey loam is generally considered as the best type of soil both for the construction of the pond and for vegetation.

D. Vegetation

Vegetation in an area can also provide insights regarding the quality of the soil. The type and density of vegetation is indicative of the productivity and physical property of the soil.

E. Topography and Drainage

Topography refers to the lay of the land with respect to the variations in elevation. Site suitability should be assessed based

on the economic cost of converting the area into a fish pond.

Generally, flat areas with a slope of about 3 percent is considered a good area for aquaculture.

F. Watershed and Climatic Considerations

Precipitation and wind are the primary climatic factors considered in evaluating the suitability of a site for aquaculture production.

Prevailing winds are considered in designing to its erosive properties and its ability to provide natural aeration to the pond.

Rainfall and watershed is also evaluated due to its influence in flooding and runoff. Generally, a larger watershed produces a larger quantity of runoff.

Ponds should be designed based on the height of a flood with a recurrence interval of 10-15 years.

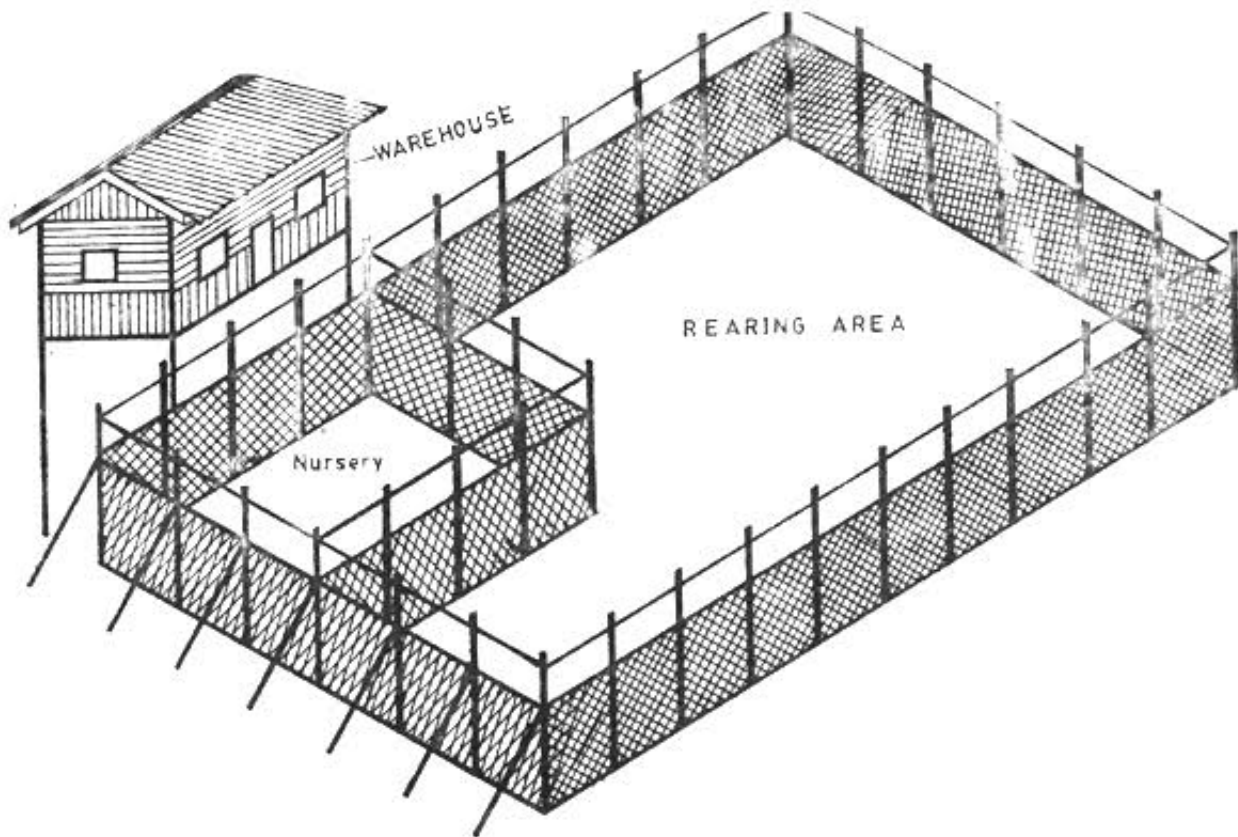
C. Aquaculture Systems

A. Pen and Cage Culture

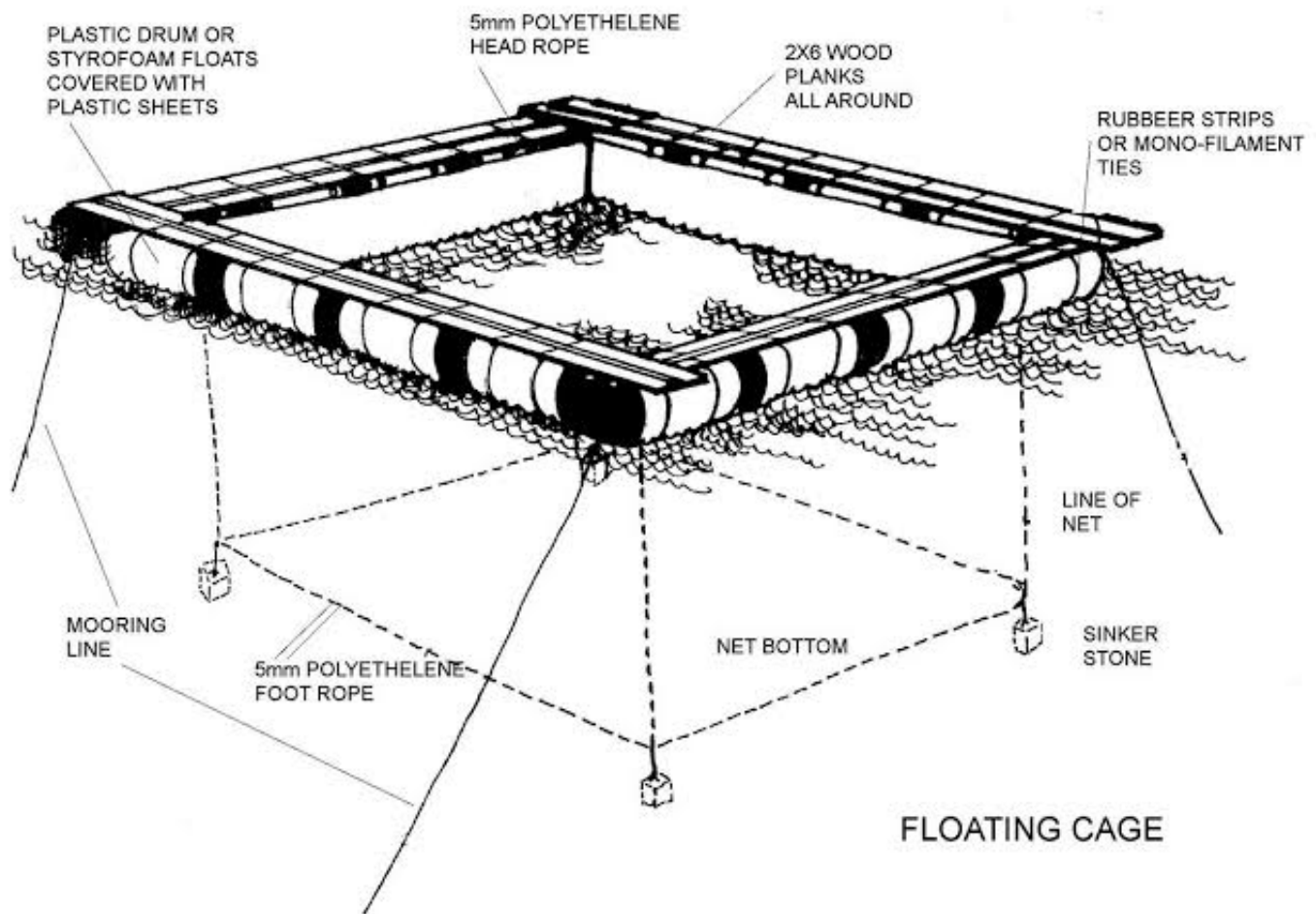
- An open enclosure system (fish are captive, free water exchange) which allow the stock to interact with the immediate environment with little to no restrictions.
- Usually constructed on public-owned natural bodies of water.

Pen	Cage
Enclosed on the sides, the bottom if formed by the lake or sea bottom	Enclosed on all sides except the top.

Fish pen



Fish cage



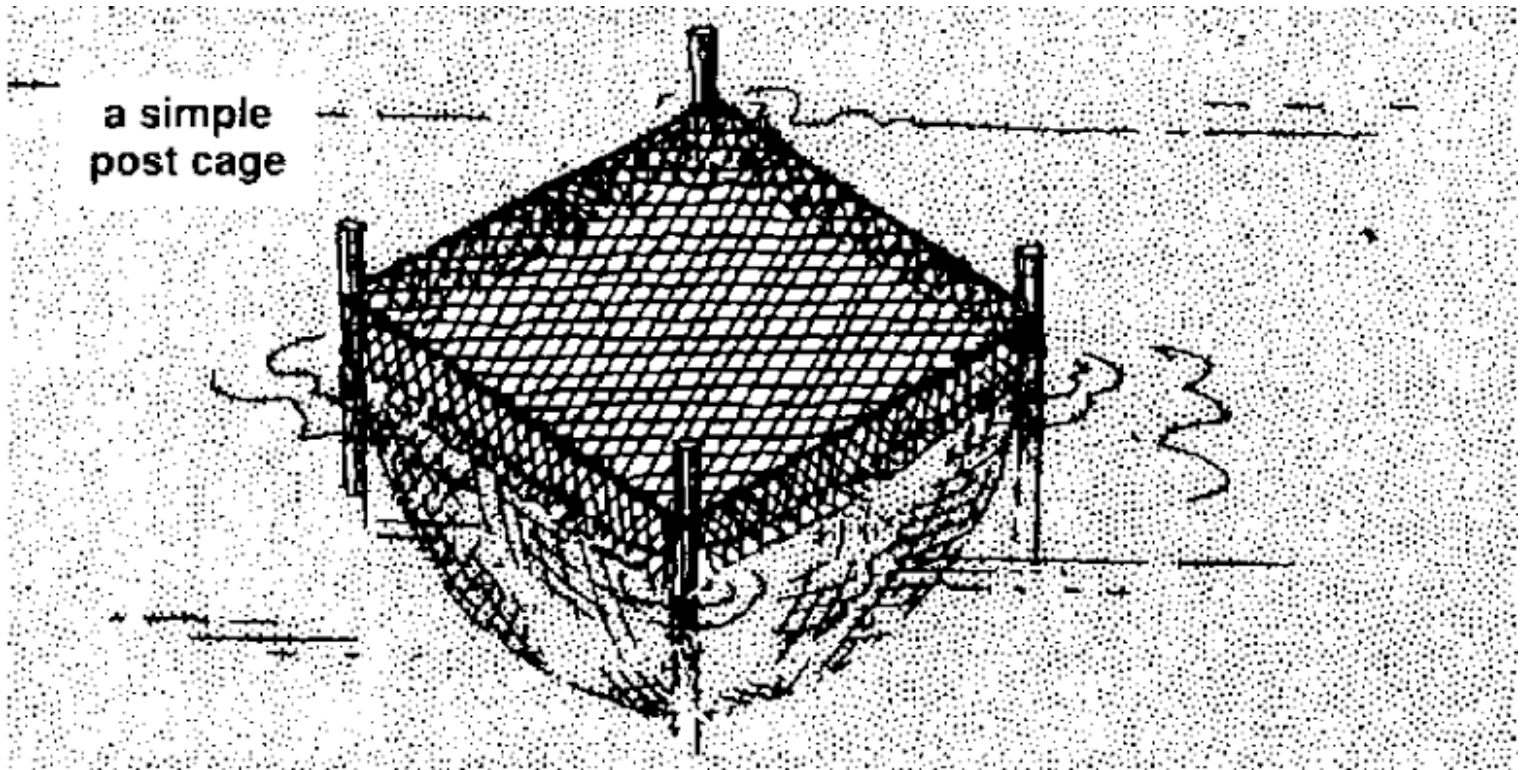
Rigid fish cage



Flexible fish cage



Fixed to bottom



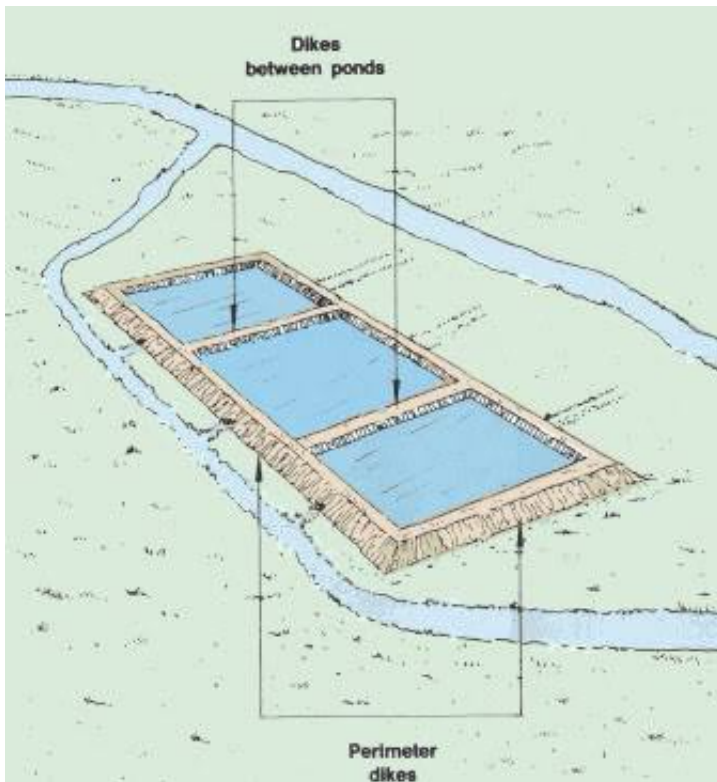
B. Ponds

a. Types of Ponds

- Watershed Ponds - involves the construction of dams in natural valleys. The depth of water is dependent on the topography of the area. the watershed is the source of water.
- Excavated / Dugout Ponds - reservoirs are dug out in the soil and is filled with water for the purpose of aquaculture production. Generally constructed in relatively flat areas and has to be pumped to drain.



- Embankment Ponds - levees, dikes, and bunds are constructed to contain water for raising aquaculture organisms. This is the primary method of pond culture and can be constructed over a wide range of topography. This process involves both digging (for depth) and construction (for embankments)



b. Considerations in Pond Construction

- Pond design should be dependent on topography, water supply, drainage, and purpose.
- Smaller ponds incur greater construction and maintenance cost but is easier to manage.
- Deeper ponds are expensive to construct.
- Oxygen depletion is less likely on larger ponds but contamination and infestation is harder to manage. It is also more prone to the erosion due to the effect of wind (produces waves).

Wave Height Formula - $h = 100 \times 1/3(\text{sqrt of fetch})$

Where fetch is the distance over which the winds blows.

Levee Design Considerations

- Perpendicular to wind reduces susceptibility to erosion.
- Parallel increases wind action and helps with aeration.
- Add 20% for core trench and 15% for settlement.

Common depth of ponds is 0.6 to 1 m

C. Raceways (Flow Through Systems)

Raceways are culture units in which water flows continuously, making a single pass through the unit before being discharged.



D. Recirculating Aquaculture Systems (RAS)

An aquaculture production system that recycles and renovates water for the culture of an aquatic organism. Self-contained aquaculture system that requires minimal water exchange due to the internal recycling of water and use of biological and mechanical filters.

